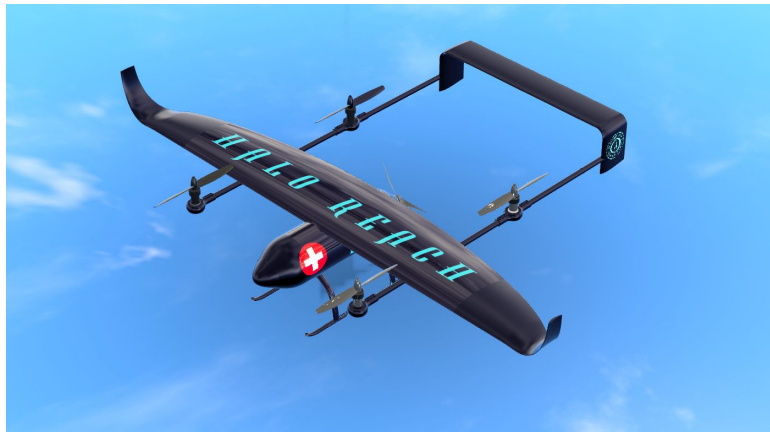

HALO REACH: On-board Avionics, FCC, Sensor Integration, HUMS and Wiring System

UAS for Disaster Relief - AE3 2024



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Abstract

Autonomous technology is revolutionising the aviation industry, and there is a great demand for this technology to be used in disaster relief efforts due to being able to perform tasks humans either can't, or which avoid risk to human life. This report presents the design on-board systems for REACH, a UAV tasked with supply delivery and communications relay missions, specifically exploring the FCC architecture, sensor processing methods and selection, the HUMS system, and wiring and shielding with a heavy emphasis on redundancy. The FCC is the heart of the FCS, taking in all data and outputting control response to actuators. REACH has a heavily redundant system consisting of 2 Pixhawk Auterion Skynode FCC's and an SCU (servo control unit). Sensor fusion is required for state estimation, and REACH uses a Decentralised Extended Kalman Filter architecture which is housed within the Nvidia Jetson AGX Xavier hardware, with the data being transmitted via I2C protocol. The benefits of RNN's is discussed, however concluded to not be trusted enough to be implemented yet. The HUMS ensures faults in all on-board components would be detected and flagged to operators, while also ensuring the battery remains cool through the use of 2 10 CFM fans. Wires are copper with silicone insulation, and selected based on the current rating of each component, resulting in a huge range of wires from AWG000 to AWG32. The whole on-board avionics system is shielded with Aluminium against EMI. The systems meet the requirements of REACH, and propose exciting prospects for future iterations.

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Nomenclature

ρ	Density of shielding material
A	Surface Area of shielding material
AWG	American Wire Gauges
CFM	Cubic Feet per minute
CG	Centre of Gravity
CNN	Central Neural Network
COTS	Commerical Off the Shelf
ECEF	Earth-Centered Earth-Fixed Coordinates
EKF	Extended Kalman Filter
EMI	Electromagnetic Interference
FCC	Flight Control Computer
FCS	Flight Control System
FLOPs	Floating Point Operations per second
GCS	Ground Control Station
GNSS	Global Navigation Satellite System
GRU	Gated Recurrent Unit
HALO	Humanitarian Aid Logistics Operations (Company Name)
Hidden Size	Number of units in GRUs hidden layer
HUMS	Health and Usage Monitoring System
IMU	Inertial Measurement Unit
Input Size	Number of features in input data at each step
INS	Inertial Navigation System
KF	Kalman Filter
LLA	Latitude, Longitude, Altitude Coordinates
m	number of measurements used to update EKF
n	number of state variables EKF is estimating
NED	North-East-Down Co-ordinates
PNT	Position, Navigation and Timing
REACH	Rapid Emergency Aid and Communications Hub (UAV name)
RNN	Recurrent Neural Network
SCU	Servo Control Unit

Sequence Length	Number of Time Steps GRU produces at once
t	Thickness of shielding material
T_{Comm}	Communication Time Delay
T_{EKF}	Time Delay from EKF
T_{GRU}	Time Delay from GRU
T_{Sensor}	Sensor Latency
T_{Total}	Total Time Delay
$T_{Transmit}$	Transmission Time Delay
UAV	Unmanned Aerial Vehicle
VO	Visual Odometry
VTOL	Vertical Take-Off and Landing
W	Weight of shielding

1 Introduction

Natural disasters have been damaging economies and livelihoods around the globe for decades. Recent years have seen an increase in frequency and severity, from fires destroying beautiful landscapes like Maui, to earthquakes causing over 50,000 deaths in the Middle East [1]. While the world must take action against the causes like climate change, there is a huge need for engineers to take action to mitigate the severity of the effects. Emerging autonomous technology is rapidly changing the landscape of the aviation industry, and can revolutionise disaster relief efforts across the world. HALO aims to be a pioneer in the new age of robotics and UAV's providing solutions beyond human capabilities.

HALO proposes a UAV that can carry out both supply delivery and communications relay missions to support disaster relief efforts. Each sub-team initially proposed a baseline configuration after evaluating all options, this is explained in further detail by Tharm [2]. Once this was selected, the systems and propulsion team focused on designing the frameworks of all systems on both the ground and UAV, and sizing the powerplant system. This report presents the first iteration of the preliminary design of the on-board subsystems and processes, with emphasis on ensuring each system has sufficient redundancy to keep the UAV operational in worst-case scenarios.

2 On-board Systems Architecture

A system is a set of interrelated components functioning together toward some common objective or purpose [3]. For REACH, the overall objective of the system is to facilitate reliable and safe carry-out of both missions while ensuring all required data is recorded and communicated between all elements of the system. The key elements are the UAV, the ground control station, the satellite, communications, the environment and other aircraft.

2.1 The on-board subsystem considerations

The onboard subsystem is responsible for recording data to both control and influence the actions of the UAV, and to communicate information back to the GCS (Ground Control Station). This involves sensors for all kinds of data, the sensor processor, the FCS, the data relay, the actuators, and all the wiring in between. The on-board system must be heavily redundant and ensure the operational safety of the UAV. The main components of the onboard subsystem are shown in Figure 11 Appendix A. The on-board system must also design against external factors that could impact the UAV. These are detailed by Patricia [4]. The ones which are considered through the on-board design are cyber threats, acidic environments from volcanoes, ash and smoke at operating altitude, and noise. The designs of the various systems detailed in this report ensure these don't affect performance.

3 Flight Control System

The Flight Control System (FCS) is responsible for receiving data from sensors on the UAV, and from the GCS, determining the desired vehicle response, and outputting to the actuators, motors and back to the GCS to achieve this response. Figure 12 in Appendix B shows an outline of how the systems sub-team work in conjunction with the control sub-team in designing the optimal FCS [5].

The Flight Control Computer (FCC) is the component generating the desired responses from the input data at the heart of the FCS. This makes it the most integral component of the on-board avionics, and so must be designed to handle all the data movement and also have sufficient redundancy to maintain safe operation of the UAV in complete signal / hardware failure scenarios.

3.1 FCC Structure

The FCC consists of 2 separate levels, the inner and outer loop. The inner loop is governed by the flight controller component of the FCC, and is responsible for stabilisation of the vehicle [6], and so

houses the control laws designed by the control team. This loop is fully autonomous and operates at a very high frequency. The outer loop is governed by the flight computer, and is essentially the decision making brain of the UAV. This processes position data, and generates set-points and desired angle / rate commands, which are sent through to the flight controller. The outer loop operates at a slower frequency and so has a less constant stream of data meaning it should not be responsible for handling stabilising actions. The outer loop also receives commands from the operations team in the GCS, and relays data back to the GCS through the data relay system [6] [7]. The FCC selected for REACH will have both a controller and computer to prevent the possibility of computer CPU overload.

3.2 FCC Case Studies

To get a better understanding of FCC configuration and redundancy, the Rascal helicopter was investigated. The Rascal FCC assembly is divided into 2 physically separate elements - the FCC and a Servo Control Unit (SCU) [8]. The FCC carries out the functions detailed in Section 3.1. The SCU, meanwhile, monitors the FCC. If the SCU detects failures resulting in unstable / catastrophic flight control transients, it reverts the aircraft to manual pilot control [9]. This adds an important extra layer of redundancy, which is something REACH should implement.

The Predator, Hunter, Global Hawk, Pioneer and Shadow are some of the most prolific UAV's in industry. Across all 5, there have been 627 mishaps per 100,000hrs of flying [10]. Failure in flight controls is the number 2 cause for these, causing 19-28% of crashes. 13% is caused by failure in the onboard avionics. Therefore it is of paramount importance that the full onboard avionics system has several layers of redundancy. As the FCC is the main component governing and connecting all the avionics together, it was deemed a necessity to have several layers of redundancy, which can come either through hardware or software.

3.3 FCC Architecture

The chosen architecture consists of 2 on-board FCC's and an SCU. As it is hard to add sufficient layers of redundancy in software, and based on the historical data discussed in Section 3.2, there needs to be a backup FCC hardware in case the first fails. This decision is supported by research conducted (Section 3.5) showing that the average COTS mass is around 200g, average size is 90x50x30mm, and average power consumption is 6W. These are all fairly insignificant constraints compared to the huge redundancy benefit from having another FCC. It was decided to have an SCU like the Rascal as a further redundancy, which also comes into effect if the HUMS detects any major faults (Section 5.1). The architecture is shown in Figure 3.

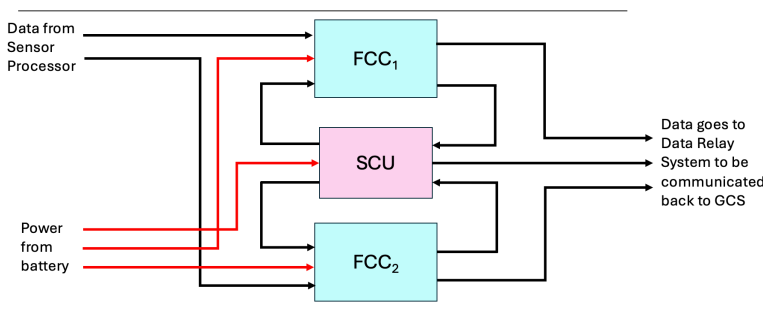


Figure 3. The Architecture of the FCC assembly

The SCU will revert the UAV to remote manual control from the GCS in certain scenarios where the SCU deems the UAV to be unrecoverable via autonomous operation. All control laws are saturated, for example, as detailed by Thilak [11], the altitude controller saturates the commanded climb gradient at ± 3 deg. This means the UAV will not receive a command exceeding this range and therefore the SCU doesn't

need to monitor control response. However abnormal feedback from the states itself, for example if a rotor or control surface stops working, can cause the UAV loses significant velocity which can be disastrous. The stall speed is 26ms^{-1} , so if the velocity hits 20ms^{-1} , the SCU will engage. Therefore the SCU will be connected to and monitor all components of the onboard avionics. The GCS receives all outer loop flight data, and will have a manual override if the operators deem it to be necessary.

3.4 External Considerations

The FCC is the most important component of the on-board system, and hence must be designed against all potential and external threats. As detailed in Section 6.3, there will be shielding on the FCC's to prevent Electromagnetic interference (EMI). The FCC's will be housed within the fuselage, which is made from TORAY prepreg 3690, as detailed in [12]. This provides protection against harsh environmental conditions like high temperatures, high acidity, and chemicals in ash clouds. Each FCC has its own wiring for power and data transmission, so the failure of one won't impact the other. The voltage and temperature is also monitored for abnormal readings, as detailed in Section 5.1, which monitor for electrical faults in the FCC's, and communicate with the SCU if manual override is needed.

3.5 FCC Hardware Selection

Due to the complex nature of the internal workings of FCC's, COTS (commercial off the shelf) hardware for the 2 FCC units are used. This increases reliability as COTS hardware has been tested and implemented in current UAVs, and also reduces cost for developing hardware from scratch. The key constraining parameters for the selected FCC's to be optimal were weighted in importance as shown in Figure 4. 7 hardware were selected for a further trade-off analysis based on these parameters as well as their suitability for the UAV. The R values were assigned based on extensive research into specs and data online [13, 14, 15, 16, 17, 18]. The 'W' columns of each of the 7 parameters show their weighting. The total of all 7 weightings is 1.

FCC	Indexes																				
	Processing Power			Mass			Size			Power Cons			Sensor Interface			Reliability			Redundancy		
	W	R	WxR	W	R	WxR	W	R	WxR	W	R	WxR	W	R	WxR	W	R	WxR	W	R	WxR
A	0.225	9	2.025	0.05	8	0.4	0.1	6	0.6	0.1	10	1	0.15	6	0.9	0.225	10	2.25	0.15	9	1.35
B	0.225	9	2.025	0.05	7	0.35	0.1	8	0.8	0.1	7	0.7	0.15	8	1.2	0.225	10	2.25	0.15	9	1.35
C	0.225	8	1.8	0.05	7	0.35	0.1	5	0.5	0.1	8	0.8	0.15	8	1.2	0.225	8	1.8	0.15	7	1.05
D	0.225	9	2.025	0.05	7	0.35	0.1	9	0.9	0.1	7	0.7	0.15	7	1.05	0.225	7	1.575	0.15	10	1.5
E	0.225	7	1.575	0.05	10	0.5	0.1	10	1	0.1	7	0.7	0.15	7	1.05	0.225	6	1.35	0.15	6	0.9
F	0.225	7	1.575	0.05	9	0.45	0.1	9	0.9	0.1	5	0.5	0.15	10	1.5	0.225	7	1.575	0.15	9	1.35
G	0.225	10	2.25	0.05	0	0	0.1	0	0	0.1	0	0	0.15	9	1.35	0.225	10	2.25	0.15	10	1.5

Figure 4. Trade-Off Analysis for FCC

FCC		WxR Total
A	UAV Navigation VECTOR-600	8.525
B	Pixhawk Auterion SkyNode	8.675
C	Cloud Cap Technology Piccolo II	7.5
D	CUAV Pixhawk V6X	8.1
E	ThePeach FCC-R1	7.075
F	A-tech SYN AvionicsMini	7.85
G	IMMC Triton	7.35

Figure 5. Results from the FCC Trade-Off Analysis

also is powerful enough for handling the CNN used in the object detection avoidance system designed by Divij [20]. On top of this, the Skynode is directly compatible with the Nvidia Jetson AGX Xavier, which is the selected sensor processor [14] detailed in Section 4.3. Therefore the Pixhawk Auterion Skynode is selected for both the main and emergency backup FCC, and Table 2 shows its' key specs.

Figure 5 shows the sum of the WxR for each of the 7 hardware, and that the UAV Navigation Vector-600 and Pixhawk Auterion SkyNode are the best choices. The main advantage of the Vector-600 over the Skynode is the lower power consumption, however the avionics system has an allocated power consumption of 1kW as explained by George [19], and so this difference is fairly negligible. Pixhawk is a brand notorious for producing FCC's which are used in many current UAV's, and

Parameter	Value
Model	Pixhawk Auterion Skynode
Dimensions (mm)	62.45x91.65x26.85
Mass (g)	209
Power Consumption (W)	5

Table 2. Specifications for the selected FCC

4 Sensor Integration

Sensors are essential for state estimation to either control the UAV, or relay data back to the ground control station. REACH has a multi-sensor system comprised of a GNSS module and receiver, an IMU, a pressure barometer, camera and LiDAR sensor. The selection process is detailed in Divij’s report [20]. The sensors receive data from the environment, which is fused to then be used for state estimation. Figure 6 shows how process of getting a vehicle response to sensor data. The control team take charge of stages 4 onwards, and therefore require the states and any time delays from the systems team.

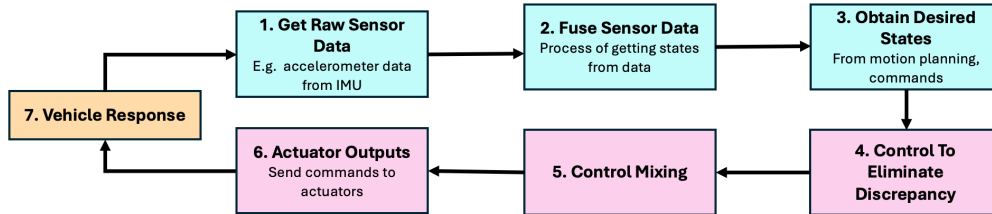


Figure 6. Sensor system

4.1 Sensor Fusion

State estimation from raw sensor data has many challenges - measurements are noisy; there are no measurements of some states; and there is data coming from several sensors. These are mitigated by sensor fusion, block 2 in Figure 6. The process improves accuracy in state estimation that could not have been possible using a single sensor alone [21]. UAV operations demand a PNT (position, navigation and timing) system that is robust [22], and so the process must be highly accurate and effective, otherwise the control system will be fed with inaccurate data leading to an undesirable vehicle response.

4.1.1 Fusion Techniques

Techniques of sensor data fusion from literature involve the use of various filters, and more recently, RNN’s (Recurrent Neural Networks). The use of low pass filter and inverting sensor model, complementary filters, Kalman and Extended Kalman Filters were investigated.

The low pass filter gives accurate estimates for body rates, attitude and airspeed, however not as good estimates for attitude and position. The slow sampling rate of 1Hz results in significant time delay inducing large errors [23] - this technique was quickly disregarded. The complementary filter consists of a low-pass and high-pass filter, each with different transfer functions, and is a computationally inexpensive fusion technique [21]. This filter is popular in UAV’s due to its simplicity. However, a complementary filter fails to estimate the gyroscope bias accurately in dynamic motion conditions, and therefore wouldn’t be as accurate as desired for fusion of the IMU data [24]. Several investigations have been conducted comparing fusion results from an IMU using a Kalman filter (KF) and a complementary filter. The mean estimation error is 6.79 times greater for the complementary filter compared to the KF [25]. The KF however does demand a more powerful micro-controller due to

the complexity. Despite this, it was decided that the Kalman approach to sensor fusion is favourable. However KF's rely on the model being linear time invariant, which in the application of REACH's operating environment, is not the case [26]. An Extended Kalman Filter (EKF) approximates the model with a linear one using a Taylor series expansion before applying a regular KF. This makes it a much more viable option than the standard KF [27], and so EKF's are used.

4.1.2 Fusion Framework

Fusion frameworks can be divided into 2 categories: Centralised and Decentralised. Centralised Frameworks are much less complex and consist of 1 master filter. As we have multiple sensors, this is not a good idea as this greatly increases the computational expense on the filter which can lead to the 'computational disaster effect' [28]. Therefore REACH will use a decentralised fusion framework, the broad outline of which is shown in Figure 7. This framework has 2 layers of filters in parallel decreasing computational cost on each filter. An added benefit is this framework introduces the possibility of adding more sensors, if REACH is to take on more mission profiles which require more sensors in the future, without any implications on data accuracy and fusion efficiency [29].

As previously discussed, EKF's will be used in the decentralised framework. A benefit of having EKF's in this configuration is that they can be used to filter out anomalous sensor data provided by vision algorithms, therefore adding further redundancy. Figure 8 shows the framework for REACH.

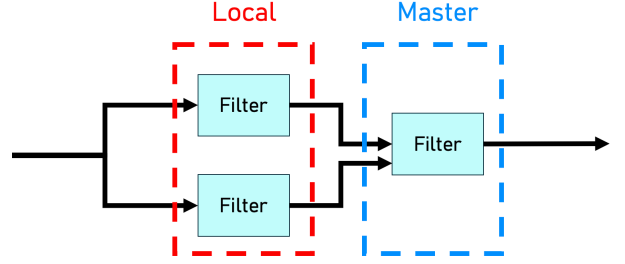


Figure 7. General Structure of a Decentralised Fusion Framework

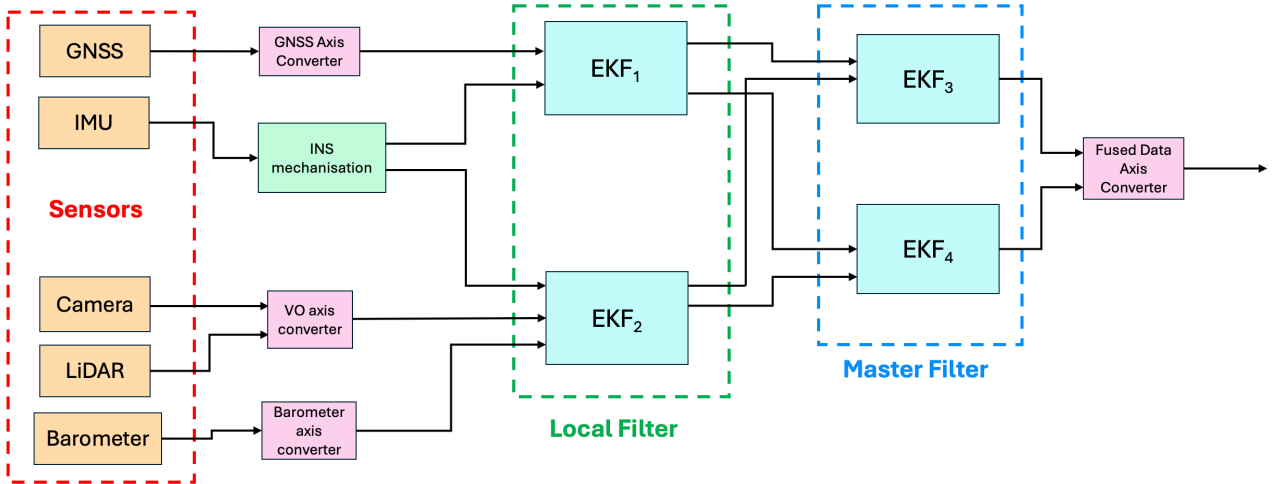


Figure 8. Fusion Architecture

We are operating in a disaster-struck environment where there will be significant external noise, and hence REACH faces the risk of GNSS outages. GNSS data is fused with IMU data due to the coupled effect from their differing frequencies. IMU's suffer from INS drift over time leading to errors in position estimation, and the fusion with GNSS data works to correct them. The process of reducing drift is known as mechanisation [22], shown in Figure 8. The INS block integrates the IMU's acceleration and angular velocity over time to estimate UAV's position, velocity, and attitude.

The fused data needs to be in the body-fixed frame for the control system, however, performing sensor fusion on data in this frame is undesirable as it heavily over complicates the process due to

constant recalculation of orientation for GNSS data [30]. The NED coordinate frame is commonly used for sensor fusion, so all data will be fused in NED coordinates and then rotated through the Euler matrix R back to body-fixed frame. The INS mechanisation process converts the IMU data to the NED axis, however the GNSS data is in the LLA axis (Latitude, Longitude, Altitude). This is converted to Earth-fixed co-ordinates (ECEF) and then to NED coordinates in the GNSS converter block in Figure 8. The barometer, camera and LiDAR data all feed into the outer loop of the FCC. The VO (visual odometry) and barometer data can be used to estimate motion from a sequence of images when GNSS data cannot be provided. Therefore they go through a separate EKF to the GNSS data to add a layer of redundancy [22]. The camera provides the N and E directional data, and the barometer measures altitude, thus providing the D data.

When designing the onboard systems of REACH, the main priority for all systems is having sufficient levels of redundancy to ensure safe operation in all environments. Therefore the master filter consists of 2 EKF's, with the second acting as a redundant filter in case something was to go wrong. The caveat of added complexity to the architecture was considered insignificant compared to the added safety margin from a second EKF.

4.1.3 Recurrent Neural Networks

Recent research has shown huge benefits from implementing Recurrent Neural Networks into fusion systems, due to their ability to cope with high-dimensional (images), and low dimensional (states) data. UAV's can face issues e.g. meteorological, that change the UAV dynamics, of which EKF's have limitations [29]. Even just operating in urban environments, where non-line of sight multi-path signal propagation can degrade the quality of GNSS data, causing an undesirable vehicle response, limits the effectiveness of EKF's. RNN's use past information and capture temporal dependencies to enhance state estimation and filter out all non-relevant noise [30].

However, one main disadvantage of implementing RNN's is the increase in computational cost and difficulty in long term data storage, which would be required for long endurance missions like that of REACH. If the weights of the RNN cell are too small, learning rate is small, and handling data is difficult resulting in the 'vanishing gradient effect' [31].

Gated Recurrent Units (GRU's) have been proven useful in mitigating these issues. The structure is much less complex than that for the conventional RNN framework, and therefore does not suffer from delays in learning rate.

4.1.4 Future Use of GRU's

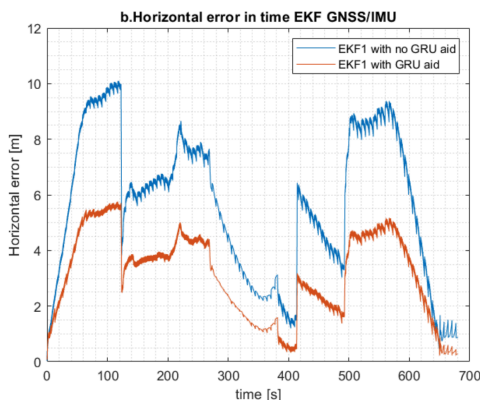


Figure 9. [29] plot showing error from GRU-EKF hybrid architecture vs EKF

Having a hybrid structure with EKF's and GRU's compared to all EKF's in both the local and master filters is explored in [29]. Implementing GRU's results in a 60% improvement in accuracy of state prediction [22]. This can be seen in Figure 9, which is the result of two architectures performing simple Vertical Take-off, cruise, and land procedure. Several papers delve into the benefits [22, 29], but the use of GRU's is still novel and not trusted yet in the autonomous aviation industry, and they require extensive training beforehand from relevant data.

Architectures with GRU's were explored, and Figure 13 Appendix C shows what the architecture would be if GRU's were implemented, with 3 GRU's each for the different NED co-ordinate directions. This can be taken even further, the whole GNS, from planning

through to control can be replaced with a neural network. Camera data is used to train the RNN, and predictions are then fused with VO through an EKF [32]. This has proved to be useful in drone racing,

where movements are very quick and agility is a very constraining metric, and so having pre-predicted states can greatly improve reaction time. This can greatly simplify and improve GNS systems in the future if they can be up-scaled to larger UAV scales.

4.2 Communication Protocol Selection

It is important to determine how data is transmitted throughout the navigation system, and all systems in general on-board the UAV [33]. There are various communication protocols which can be adopted for internal data transmission. The most commonly seen protocols are detailed in Table 3.

Protocol	Transmission Speed (Mbps)	Use with Sensors?	Historical use in UAVs?
UART	0.0096 - 0.1152	Not Common	No
SPI	10-20	Yes	Not Common
I2C	0.1-0.4	Yes	Yes

Table 3. Comparison of Communication Protocols

Table 3 shows SPI has the fastest transmission speed, however isn't commonly used on UAV's as it requires more power. I2C has a transmission speed that is deemed fast enough for use on REACH, and requires much less power, therefore I2C was selected.

4.3 Processor Hardware Selection

The Sensor Processor Hardware houses the fusion architecture determined previously. It is responsible for receiving the data from the sensors, carrying out the fusion, and outputting the estimated states to the FCC. A similar selection process as that for the FCC was conducted, where the main constraints were the same. This investigation resulted in the Nvidia Jetson AGX Xavier [34] being the optimal hardware for REACH. This is also directly compatible with the Auterion Skynode FCC's [14]. Table 4 shows the specs of the selected hardware - FLOPs is floating point operations per second. This parameter is used in the time delay calculations in Section 4.4.

Parameter	Value
Model	Nvidia Jetson AGX Xavier
Dimensions (mm)	100x87x20
Mass (g)	300
Power Consumption (W)	10-30
FLOPs	10×10^{12}

Table 4. Specifications for the selected sensor processor hardware

4.4 Time Delay

A major concern when an architecture is selected is the time delay from the point the data is measured to the point that the state estimations are implemented into the controllers. Equation 1 shows the components that are considered for the time delay calculation.

$$T_{Total} = T_{Sensor} + T_{Comm} + 2T_{EKF} + T_{Transmit} \quad (1)$$

T_{Sensor} is the time for the sensor to detect the event and convert readings to an electrical signal, which is found on the sensor specifications. T_{Comm} is the time it takes to transmit the data to the processor, and is dependent on the selected communication protocol. T_{EKF} is the time taken for each layer of EKF's to process the data. The actual estimation of this involves calculating time for state prediction, jacobian calculation, and measurement update [35]. However as the values are very small,

this is deemed too excessive, and therefore a grossly approximated method is used to get an estimate for T_{EKF} shown in Equation 2 where n is the number of inputted states, and m is the number of update measurements [26].

$$T_{EKF} \approx \frac{2n^3 + 2nm^2 + m^3}{\text{Hardware FLOPs}} \quad (2)$$

For each of the calculations, the max value is used as well as a large buffer due to approximations. This resulted in a total time delay of 0.008s for the architecture, which is rounded to 0.01s for redundancy. This was deemed acceptable by the control team, verifying the selected architecture does not pose extensive time delays ensuring a quick vehicle response.

The time delay for the GRU hybrid architecture shown in Figure 13 Appendix C was also calculated to see if there was a difference if this architecture was to be used in the future. The change in the calculation is $2T_{EKF}$ becomes $T_{EKF} + T_{GRU}$. A similar grossly approximate method is used for T_{GRU} due to the complexity of modelling the update and reset gates, and candidate activation time. This is shown in Equation 4 in Appendix C. The T_{GRU} time delay is 3 orders of magnitude quicker than the T_{EKF} . Despite this, this gave a time delay of 0.0079s, showing that the change in time delay overall is negligible from implementing a GRU hybrid structure.

4.5 Finalised Selection

For now, REACH will use the EKF decentralised fusion architecture shown in Figure 8, as the low reliability of GRU's due to lack of current commercial use vastly outweighs the slight benefits of more accurate estimations in quicker time. The benefits of having pre-estimated data from RNN's in the case of signal failures and GNSS outages is mitigated in the chosen architecture by the use of INS mechanisation, and redundant filters. A key reason for using RNN's is their effect on mission profiles that require high agility. Currently, our mission profiles don't require this level of precision.

4.6 Navigation System Placement

The positioning of the components of the navigation system has many constraints as well as being very important for weight and balance. Figure 10 shows the layout of the system on the fuselage.

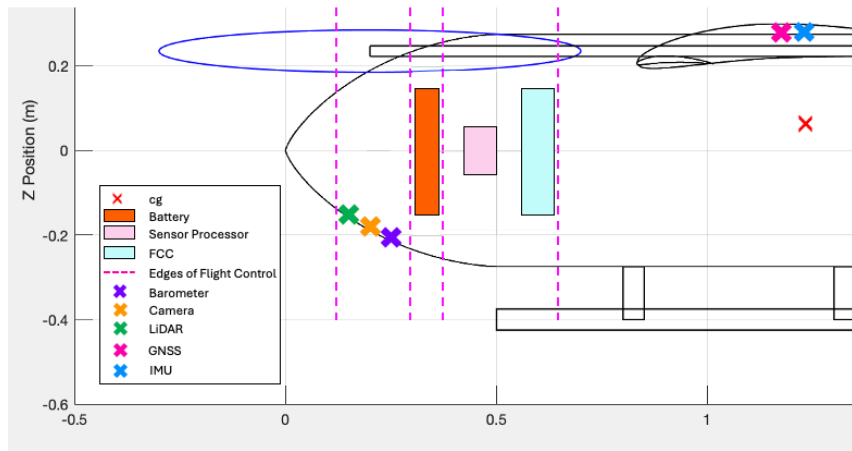


Figure 10. Navigation System Placement on UAV

The battery is housed between the sensors and the area where the the Nvidia Jetson sensor processor and the Auterion Skynode FCC's are housed. All sensors need to be positioned away from the motors

and propellers to limit interference and vibrations affecting data. In Figure 10, it can be seen that all sensors are located far from the push propeller at the rear end of the fuselage, and are on the centre-line of the fuselage body to be far enough from the VTOL propellers. Each sensor has further constraints. The GNSS needs to be placed on top of the UAV to ensure a clear signal between the receiver and satellites [36]. The IMU needs to be placed at the cg of the UAV as it is estimating /measuring the UAV's motion and attitude, and don't want rotational effects on the data from off-centre placement [37]. The pressure barometer should be on the underside of the UAV to be shielded from rain and to be far away from propeller wash causing turbulence [38]. The camera needs unobstructed Field Of View of whats underneath the UAV, and also in front of it [39]. It will incorporate a tilting feature to accommodate this. Finally the LiDAR sensor also needs a clear 360 field of view to maximise scanning for obstacle avoidance [40]. Therefore it is placed at the very front on the underside of the UAV

5 Environment and Monitoring System

5.1 Health and Usage Monitoring System

The HUMS is responsible for recording the status of the critical systems onboard the UAV for early detection of any defects, thus rectifying them or returning the UAV safely before having catastrophic implications [41]. All HUMS data must be relayed back to the GCS, so operators can intervene at any point deemed necessary. The SCU, detailed in Section 3.3 therefore plays an integral role in the HUMS by monitoring the data going through the FCC and reverting control to the remote pilot. The main components to be monitored are show in Table 5 with their respective selected fault detection methods [42]. The table also shows key environmental conditions that should be monitored.

Component	Fault Detection Method
Sensors	EKF's in sensor fusion and potentially GRU's in future
Engine	Vibration monitoring through accelerometers, temperature monitoring through thermocouples, fuel pressure through a pressure gauge
Battery	Voltage monitoring through the power board outputting to the various electronics, thermistors for temperature monitoring
Propellers	Optical sensors for RPM, vibration monitoring through accelerometers on propeller hub
Electronics	Voltage and temperature monitoring at regular intervals
Airframe	Strain Gauges to detect deformation, accelerometers for vibrations
FCC	Voltage and Temperature monitoring
Actuators	Strain gauges for torque and force measurements, potentiometers for position
Acidity of Environment	pH sensor

Table 5. Fault Detection methods for key UAV components

It can be seen that measuring temperature, vibrations, strain, and voltage are the key techniques used in the HUMS. In the next iteration, after several field tests, each component and sensor will be given an operating bracket with a max and min value for the readings of vibrations etc, and will be flagged up to the GCS if this is exceeded. If the operators deem the fault to be too great, a protocol will be initiated to land the UAV at the closest safe location, however if deemed to be rectifiable by onboard control, the UAV will remain operational. The specific design of this protocol will be designed in future iterations.

5.2 Propulsion Cooling System

The battery selected is the Amprius SIMAXX 400 Wh/kg, as detailed by Tharm [2]. This has an operating temperature range of -30°C to 55°C. As we are only operating at a max altitude of 500m,

falling below the minimum temperature is not a threat to REACH, and therefore no heating system is required. However, avionics heating and volcanic / wildfire environments can result in the maximum temperature being exceeded. Therefore REACH will have a cooling system to keep the battery below 45°C at all times - a 10°C buffer. This is also monitored, as detailed in Section 5.1. The battery has a discharge rate of 10C, meaning the battery can discharge at ten times its capacity, which can greatly increase temperatures [43], and therefore active cooling is required as opposed to passive. The three cooling methods investigated for REACH were liquid cooling, fans, and heat sinks, as these methods were regarded sufficient enough to keep the battery below 45°C after extensive research. Fans were selected for REACH due to being very light in comparison, and weight is a big constraint for REACH. 2 fans, each with a 10 CFM rating can dissipate 35W of heat, which is adequate for the selected battery, and adds an additional weight of only 60g [44]. In future iterations of REACH, further cooling systems can be explored and the exact fans will be selected, especially if REACH is to expand its mission profile and operate in more adverse environments.

6 Wiring and Shielding System

The wiring system connects all the on-board system components and consists of the wires within the flight control system transmitting data, and the wires supplying power to all the on-board components.

6.1 Wiring Selection

The wires were selected through a universal wiring method to ensures ease of getting COTS wires - the standard American Wire Gauges (AWG) [45]. The wire gauge is a measurement of wire diameter. All wires are copper with silicone insulation protecting the copper. The data transmission wires must support the I2C communication protocol, and therefore AWG 28 is used [33]. The wires supplying the power to the onboard components require a calculation of current draw rating dependent on each component. These were obtained from data in the specs for each component. From this, the required AWG of the wires for each component was determined using the chart in [46]. Table 6 shows the selected wires. In each case, the max current draw was taken from the specs of each component, referenced in their respective selection processes in reports [2, 20, 47, 19] or calculated from the current voltage power relation. Each AWG selected has a max current exceeding these values. This is to ensure there is no loss of power through increased resistance causing thermal losses and overheating.

6.2 Wiring Weight Estimation

In order to get an estimation of the weight of whole wiring system, the lengths of each wire is needed. These were estimated from the whole UAV layout and weight and balance data, which is presented in [2]. The weights by per unit metre of each AWG of wire used were taken from [REF], and then multiplied by the estimated lengths for each component. Table 6 shows the results of the main components of the navigation, communication, and propulsion systems with the total wiring weight coming to 1.978kg. The CG of the wiring system is very hard to calculate, however upon rough estimations, it can be seen that it is very close to the cg of the UAV due to the motor wires being the heaviest at the back of the fuselage, and the rest of the wires all at the front. Therefore the cg of the wiring system is the same as that of the UAV.

Component	Current Draw (A)	Wire needed	No. components	Weight (kg)
Motor	280.00	AWG 0000	4	1.36
FCC	5.00	AWG 14	2	0.0299
Camera	2.40	AWG 17	1	0.00700
IMU	0.10	AWG 32	1	0.000638
GNSS	1.80	AWG 19	1	0.0114
LiDAR	0.83	AWG 22	1	0.00220
Sensor Processor	3.33	AWG 16	1	0.0063 0
Actuator Processor	4.00	AWG 15	2	0.139
Data Relay	0.80	AWG 22	1	0.00470
Comm System	10.40	AWG 11	1	0.0200
Generator	35.72	AWG 6	2	0.338

Table 6. Power wire selection and weights

6.3 Protection of Avionics

As stated in Section 2.1, there are several environmental and external concerns that can result in failure of avionic especially EMI. REACH has shielding on the main avionic components and wiring. EMI is a disturbance interrupting and degrading performance of avionics, and can come from (un)intentional jamming or congested radio frequency backgrounds [48]. To mitigate against this REACH will use a twisted pair shielded wires configuration [49]. The twisting mitigates EMI interference and interference in general, therefore reducing noise. The most common materials used for shielding against EMI are aluminium and copper. Copper has better conductivity, however is much heavier and more costly. Aluminium has the best strength-to-weight ratio, however can form an oxide over time resulting in material degradation [50]. As there are pros and cons to both, weights using both copper and aluminium were calculated.

$$W = \rho t A \quad (3)$$

To get a weight estimation of the shielding, Equation 3 was for each component and wire, where ρ is the density of the material, A is the surface area, and t is the thickness. Table 7 shows the results. The thicknesses were obtained from [51], and densities from [52].

Parameter	Copper	Aluminium
Density (kgm^{-3})	8940	2712
Thickness needed (m)	0.003	0.005
Weight of hardwares (kg)	3.0690	1.547
Weight of power wires (kg)	2.222	1.124
Weight of data wires (kg)	0.9195	0.4649
Total Weight (kg)	6.666	3.370

Table 7. Weights of shielding for different materials

Due to limited availability in weight for the avionics and wiring systems, after consultation with weight and balance (detailed in [2], the copper shielding was deemed too heavy, and so the lighter aluminium was favourable. There is the issue of oxidation, however this is mitigated by applying a layer of paint to all the aluminium shielding.

7 Conclusions and Reflections

This report presents the first iteration of the preliminary design of some of the key on-board systems of REACH. The main focus of the design of this iteration was to have sufficient redundancy to keep the UAV operational in all conditions, which was implemented where possible throughout.

7.1 Flight Control System

The architecture selected is a result of extensive market analysis of COTS hardwares, with emphasis on brand reliability, power and redundancy. It effectively enables REACH to process all data seamlessly for optimal control. The 2 FCC and SCU architecture is novel in the autonomous industry making it one of the most reliable UAV's. The next iteration would involve designing the internals of the SCU and how it will communicate with the GCS for the pilot override, and after conducting field tests, look further into selecting a more optimal processor board within the FCC.

7.2 Sensor Integration

The navigation system is designed such that if REACH is to take on more missions in the future, the relevant sensors can be added easily. The selected fusion architecture is optimally designed based on trusted methods in the industry, with added redundancy. However as described in the discussion of RNN's, the future is very exciting with the possibility of incorporating machine learning through robust training of GRU's. We plan for REACH to be a pioneering UAV in the autonomous flight industry, and after numerous field tests and simulations are conducted verifying the reliability of RNN's, REACH can become one of the first wide scale UAV's to incorporate such technology. Being able to predict states will enable REACH to operate in highly urban environments where agility is more of a constraining design metric, greatly shorten processing time, and enable REACH to successfully operate in environments where there could be no GNSS signal at all.

7.3 Environment and Monitoring System

The HUMS was designed such that all components of which the failure could result in disaster, were monitored for identified parameters that could go wrong. As stated in Section 5.1, the next stage of design would be to establish the operating bracket for parameter, and to select the hardwares for the various sensors, as due to time, this couldn't be completed in this iteration. The selection of the fans for the cooling system will be selected now that the type of fan needed has been identified. The result would be a fully integrated HUMS covering all possible causes of faults and relaying the data back to the GCS, who would have a robust protocol should a fault be deemed too large.

7.4 Wiring and Shielding System

Each wire was selected to support more than the current required for the appliance the wire is connected, ensuring a reliable wiring network. The silicone insulation as well as the aluminium shielding ensure no external threats can interfere with the transmission through the wires. In future iterations, a sophisticated power board will be designed to add further redundancy ensuring the right current is going through the wires, and the possibility of redundant wires will be explored in more detail.

7.5 Overall

Overall, the final design meets all requirements defined at the start of the process by all stakeholders, including the coordinators, other sub-teams, and the systems and propulsion sub-team. The combination of current trusted COTS hardware as well as novel software and hardware ideas sets up REACH to easily adapt to the exciting future of autonomous technology.

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Appendix

A On-board architecture

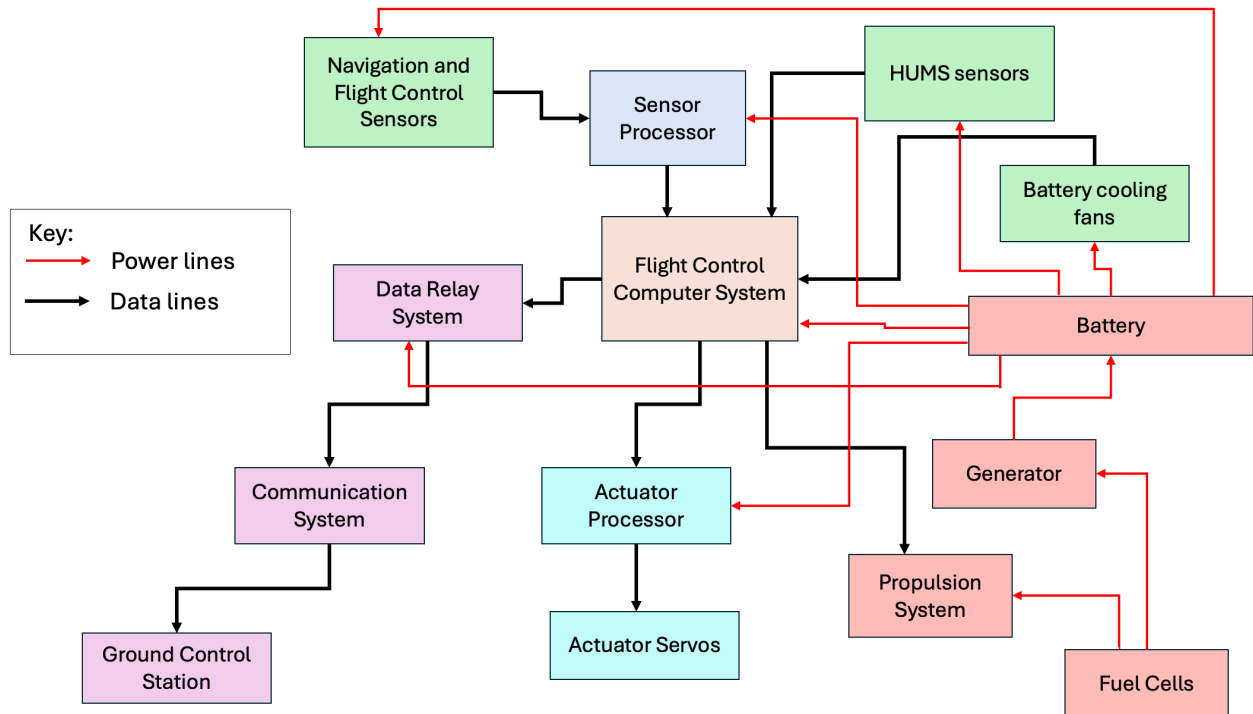


Figure 11. Architecture of the onboard system

B Flight Control System

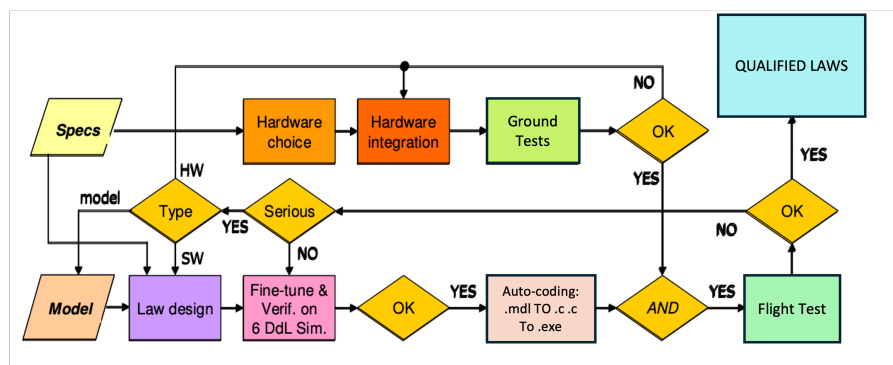


Figure 12. Flight Control System Design Process

C Incorporation of RNN's through GRU's

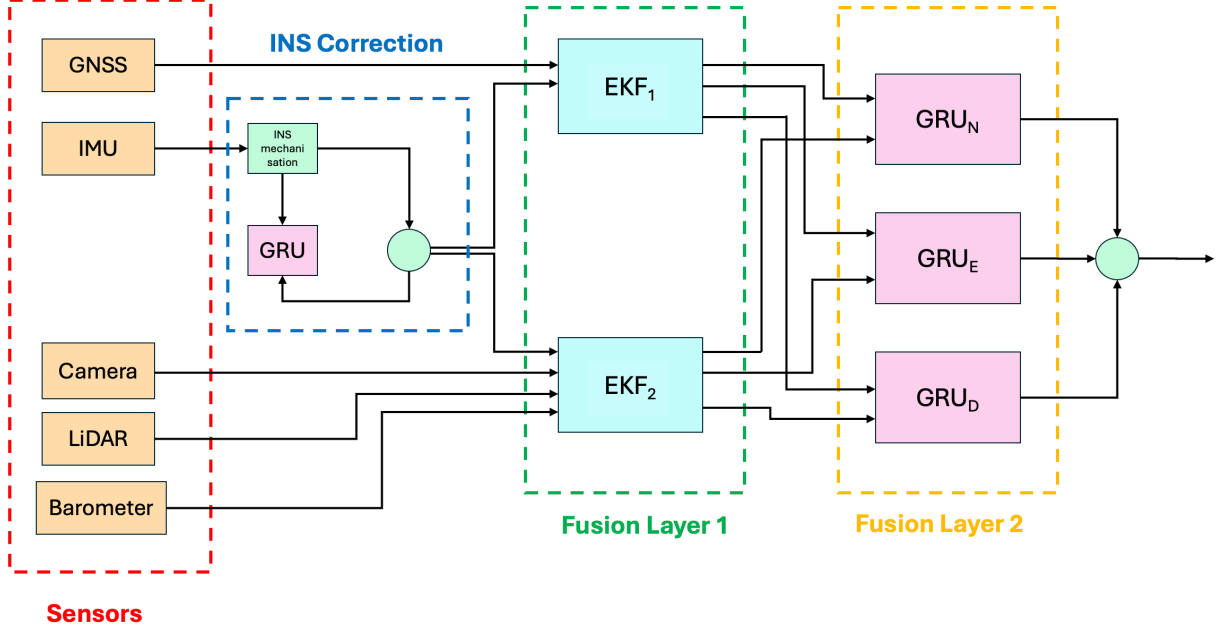


Figure 13. GRU and EKF Hybrid Fusion Architecture

$$T_{GRU} \approx \frac{6 \times sequence\ length \times (input\ size \times (hidden\ size + hidden\ size^2))}{Hardware\ FLOPs} \quad (4)$$